

AI Agents

Tools, skills, memory, and autonomous action

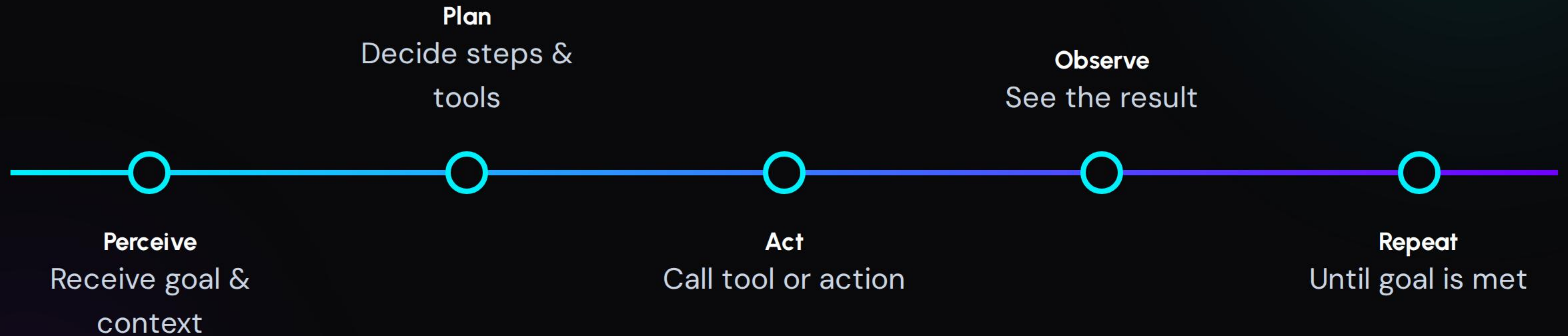
| What is an AI Agent?

An **AI agent** is an LLM that can take actions — not just respond.

- 🧠 **Standard LLM:** Prompt → Response → Done
- 🤖 **Agent:** Goal → Plan → Use Tools → Act → Report
- 🔄 Ability to loop and retry until objective met
- ⚡ Key difference: **Agency** (acting in the world)



| The Agent Loop



| Tools: Interaction Primitives



Search

Web browsing or database retrieval



Code

Writing and executing python/JS



API

Slack, Email, CRM connections



File System

Reading and writing local files



Browser

Navigating web pages directly



Math

Reliable arithmetic processing

| Skills: Advanced Workflows



What are Skills?

Higher-level capabilities composed of multiple tools and logic.

Example: **Search + Summarize + Format** = "Topic Research" skill.



Implementation

Claude Code: /commit or /review-pr are built-in skills.

n8n: Sub-workflows used by agent nodes act as skills.

Tools are primitives. Skills are the workflows built on top.

Memory Types

Type	Description	Example
In-context	Text in current prompt window	Chat history
External	Database outside the model	Vector store, SQL
Episodic	Logs of past actions/outcomes	"Last time X failed"
Semantic	Stored facts and knowledge	User preferences

| Planning Strategies

⚙️ **ReAct:** Synergizing Reason + Act (think, act, observe, repeat).

📖 **Plan-and-Execute:** Full strategy up front, then execution.

🌳 **Tree of Thought:** Exploring multiple paths in parallel.

🕶️ **Reflection:** Self-critique of output before completion.

Most production agents use **ReAct** — it's simple, transparent, and robust.

| Multi-Agent Systems



Splitting complex tasks across specialized agents:

Orchestrator: The manager who delegates.

Specialists: Researcher, Coder, Writer.

Critic: Quality assurance and reviews.

"Like a human team, each expert in their lane, coordinated by a lead."

| Defining Success



Clear Goal

Avoid wandering behavior with precise instructions.



Right Tools

Capability is limited by the available toolset.



Memory

Persistent context across multiple steps.



Error Handling

Recovery logic for when tools fail.



Stop Conditions

Knowing when to stop and report back.

| Architecture Summary



The Brain

LLM reasoning & planning engine.



Tools

Interface to the external world.



Memory

Context and history storage.



The Loop

Iterative execution cycle.

Agents in Action

From autonomous coding assistants to complex business automation, AI Agents are redefining workflow efficiency.

> **Claude Code:** Terminal-based coding agent.

🔗 **n8n:** Visual agentic workflows.



Questions?

Exploring the future of autonomous systems.

```
> agent.status = "Task Completed"
```


| Image Sources



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